

Sri Manaswi Tipparaju

Site Reliability Engineer • Cloud Platform Engineering • Infrastructure Automation

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SUMMARY

13+ years embedded inside two of the largest enterprise engineering organizations in the industry — Oracle and Cisco — keeping mission-critical platforms alive, fast, and scalable. Background sits at the intersection of performance engineering, cloud infrastructure, and automation: diagnosing production incidents, building the tooling that prevents recurrence, and running reliability programs across 20+ engineering teams at scale.

At Oracle, responsible for SLO/SLA targets on OCI-hosted cloud management platforms — building automated provisioning pipelines, observability setups, and the blameless post-mortem culture that actually reduces repeat incidents. Equally comfortable in the terminal writing Python and shell automation, triaging a P1 in a war room, or designing containerized architectures on multi-tenant OCI workloads. Recent work extends into AI-powered tooling — PerfAgent is an autonomous diagnostic pipeline that classifies performance artifacts, runs parallel LLM-based analysis across 9 diagnostic domains, and routes severity-ranked findings to the right team in under 60 seconds.

AWS, Microsoft, NVIDIA, and Linux Foundation certified with strong foundation in OCI, AWS, Oracle Autonomous DB, containerization, CI/CD and performance diagnostics (AWR, ADDM, AppDynamics, CloudWatch).

TECHNICAL SKILLS

SRE / Reliability	SLOs · SLAs · Error Budgets · Incident Management · On-Call Runbooks · Chaos Engineering · Capacity Planning · Toil Reduction
Cloud	OCI · AWS (EC2, EKS, RDS, S3, VPC, IAM, Lambda, CloudWatch) · Multi-cloud Architecture
Containers / IaC	Kubernetes (CKA) · Docker · Helm · Terraform · CloudFormation · CI/CD (CodePipeline, CodeBuild, Git)
Observability	CloudWatch · AppDynamics · OEM · AWR/ADDM/SQL Monitor · Alerting & Dashboarding · Custom Prometheus metrics
Automation / Dev	Python · Shell · SQL/PL/SQL · TypeScript · Node.js · React · REST APIs · AI/LLM Integration
Databases	Oracle Autonomous DB · Oracle RDBMS (CDB/PDB Multitenant) · PostgreSQL · AWS RDS · Aurora
AI/LLM	Agentic AI / Multi-agent Systems · Agent Architecture & Memory · NVIDIA NeMo / NIM Microservices · Agent Evaluation & Tuning · RAG (NeMo Retriever, embeddings, Pinecone) · Anthropic Claude API · Azure OpenAI / AI Foundry · Responsible AI / NeMo Guardrails · LLMops & Observability

EXPERIENCE

Site Reliability / Cloud Platform Engineer | Oracle Inc — Oracle Management Cloud & Enterprise Manager | Mar 2017 – April 2026

- Owned SLO/SLA reliability for OCI-hosted Oracle Management Cloud across multi-region deployments, building automated health checks and failover runbooks that held availability above 99.9%.
- Rebuilt the P1 alerting pipeline with structured triage playbooks, cutting MTTR by ~40% and ending the constant dashboard-hopping during incidents.
- Automated OCI database provisioning with Python and shell scripts, taking it from a manual 4–6 hour task to under 30 minutes with consistent parity across dev, staging, and production.
- Ran SQL/PL/SQL performance reviews for 20+ engineering teams, using AWR, ADDM, and SQL Monitor to clear p99 latency bottlenecks and improve query throughput by 60%+ on the busiest workloads.
- Tuned WebLogic thread and connection pools on Enterprise Manager services, improving response times by ~30% and resolving recurring SLA breaches.
- Migrated Oracle Enterprise Manager from Solaris to Linux for a Fortune 10 customer using Data Pump, completing the move with zero data loss and no downtime.
- Configured Data Guard and GoldenGate standby and replication setups to keep critical OCI workloads resilient and meet near-zero RPO/RTO targets.
- Introduced blameless post-mortems and PIR processes that reduced repeat incidents and built a shared knowledge base across teams.
- Integrated OpenAI Codex into the CI workflow to auto-generate test cases and assist code review, expanding coverage and catching issues earlier in the Jenkins pipeline.

Database Performance & Reliability Engineer | Cisco Systems Inc | Sept 2012 – Mar 2017

- Kept 200+ production databases healthy for mission-critical Cisco applications — proactive monitoring, automated alerting, and fast triage when things went sideways.
- Built a performance snapshot framework that proactively detected inefficient SQL across hundreds of DB instances before they caused incidents, shifting the team from reactive firefighting to proactive tuning.
- Automated load testing with LoadRunner before every major release to catch regressions before they hit production; diagnosed bottlenecks using AWR, ADDM, SQL Trace, and TKPROF.
- Used AppDynamics for heap analysis and memory diagnostics to track down cascading performance issues impacting end users.

PROJECTS

OpsMind (formerly PerfAgent) — AI SRE Copilot Platform [Live Demo](#) · [GitHub](#)

- Built an autonomous AI SRE copilot (Node.js + TypeScript + Anthropic claude-sonnet-4.5) that ingests production diagnostic artifacts — AWR reports, SQL Monitor output, thread/heap dumps, JFR recordings, JMeter results, browser logs, stack traces, and **Kubernetes diagnostics** — classifies them, runs parallel domain-specific LLM analysis across **10 specialized skills**, correlates findings across artifacts into a unified causal chain, and dispatches severity-ranked output to Slack, Jira, and email in under 40 seconds.
- **Multi-agent architecture (5 agents)**: Classifier (skill routing from content preview), Skill Analyzer (parallel Claude calls with 4–8 concurrency), Cross-Artifact Correlator (builds incident timeline + causal chain linking findings across files), Chat Agent (conversational follow-up on incidents), Reporter (severity-ranked notifications). Orchestrator coordinates the 6-step pipeline with Critical-severity override to #incidents channel.
- **RAG over operational runbooks**: Pinecone vector DB with OpenAI text-embedding-3-small (1536d) retrieves relevant runbooks and postmortems before every skill analysis. In-memory cosine-similarity fallback enables offline demo with zero external dependencies. Every analysis cites source runbooks in the UI.
- **LLMOps observability**: Prometheus instrumentation tracks every Claude call — input/output tokens per skill, P50/P90/P99 latency histograms, cost in USD (\$3/M input, \$15/M output), parse success rate, RAG retrieval latency. Pre-built Grafana dashboard ships in the docker-compose stack.
- **10 diagnostic skills** as structured system prompts with strict JSON output contracts — each finding requires an **evidence** field grounded in specific numbers from the input, making hallucinations immediately identifiable. Added Kubernetes skill for OOMKilled, CrashLoopBackOff, HPA failures, and resource limit analysis.
- **Production scale**: Stateless Express server on Docker / AWS ECS Fargate, robust JSON parser with 4-layer fallback (parse → repair → sanitize → regex extract), streaming responses for skill analysis, REST endpoints for `/analyze`, `/chat`, `/rag/ingest`, `/metrics`, `/webhooks/{pagerduty,opsgenie,grafana}`, `/ci/sql-gate`. Config-driven team routing — adding a new Slack channel or Jira project is one config edit.
- **Performance**: End-to-end triage of 5 diagnostic files in 25–40 seconds at ~\$0.10 per incident (vs. 2–4 hours and ~\$500 in senior engineer time for manual triage — a 200× time and 5,000× cost reduction).
- **Tech**: Node.js · TypeScript · Anthropic Claude · Pinecone · OpenAI Embeddings · Express · Prometheus · Grafana · Docker · AWS ECS · Next.js (portfolio integration)

AI-Powered Cloud Insights Dashboard [Live Demo](#)

- Built a full-stack observability dashboard with React, Next.js, and Groq LLM APIs that surfaces real-time operational insights — summaries, key signals, and recommended actions — alongside raw data.
- Designed a modular API layer with structured response rendering so the output (summary, key points, actions) is machine-readable and easy to pipe into alerting or runbook workflows.

CERTIFICATIONS & EDUCATION

Certifications: [AWS Solutions Architect – Associate](#) (exp. Apr 2029) • [Certified Kubernetes Administrator \(CKA\)](#) (exp. Apr 2028) • [Microsoft Certified: Azure AI Apps and Agents Developer Associate](#) (exp. Jun 2027) • [NVIDIA Certified Professional: Agentic AI](#) (exp. Jun 2028)

Education: M.S. Engineering, Lamar University (2012) • B.E. Engineering, Acharya Nagarjuna University